



Green University of Bangladesh
Department of Computer Science and Engineering (CSE)
Faculty of Sciences and Engineering
Semester: (Spring, Year:2024), B.Sc. in CSE (Day)

PROJECT REPORT

Course Title: Cyber Security
Course Code: CSE 323 **Section:** 212 D4

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Project Proposal Status

Marks:

Signature:

Comments:

Date:

Introduction

RSA stands for Rivest-Shamir-Adleman, which is a public-key cryptosystem invented in 1977. It's one of the most widely used for secure communication on the internet. Unlike symmetric cryptography where a single key is used for both encryption and decryption, RSA utilizes a unique two-key system: a public key and a private key.

Objective

The objective of this project is to implement the RSA algorithm in a web-based application, allowing users to input prime numbers (p and q) along with a plaintext message. The application calculates the public and private keys, encrypts the message, and then decrypts it, demonstrating the RSA encryption and decryption process.

Here's a deeper dive into the theory behind RSA:

The Math Behind RSA:

The core concept of RSA relies on the mathematical difficulty of factoring large prime numbers. Prime numbers are whole numbers greater than 1 that are only divisible by 1 and itself. Factoring a large prime number into its prime components is computationally expensive. RSA exploits this complexity to establish a secure communication channel.

Key Generation:

- Selecting Large Prime Numbers:** The process begins with generating two large, secret prime numbers, p and q . These primes are chosen meticulously as the security of RSA hinges on their size. The larger the primes, the harder it becomes to factorize the resulting value derived from them.
- Calculating the Modulus (n):** We compute n by multiplying the chosen primes p and q . This value n acts as the foundation for both the public and private keys.
- Public Exponent (e):** An integer value, e , is selected such that it's relatively prime to a mathematical function of p and q , typically denoted as $\phi(n)$. Being relatively prime implies that e and $\phi(n)$ share no common factors except 1. This value, e , forms part of the public key.

4. **Private Exponent (d):** Using a mathematical operation called modular multiplicative inverse, we calculate the private key exponent, d. This value has a special property: when e and d are multiplied modulo n (meaning the remainder after division by n), the result is 1.

Public Key Distribution:

The public key consists of the modulus (n) and the public exponent (e). This information is freely distributed, often embedded in certificates or published on websites. Anyone can encrypt messages using this public key.

Implementation

The implementation is done using HTML and JavaScript. The HTML page provides input fields for users to enter the prime numbers ('p' and 'q') and the plaintext message ('msg'). Upon clicking the "Apply RSA" button, the RSA function is triggered.

In the RSA function:

- User inputs ('p', 'q', and 'msg') are retrieved and validated to ensure they meet the requirements.
- Public and private keys ('n', 'e', 'd') are calculated based on the provided prime numbers ('p' and 'q').
- Encryption and decryption operations are performed using the calculated keys.
- The resulting ciphertext and plaintext are displayed on the webpage.

Features

- ➔ Input fields for prime numbers ('p' and 'q') and plaintext message ('msg') with input validation.
- ➔ Calculation and display of public key ('n', 'e') and private key ('d').
- ➔ Encryption and decryption of messages using the RSA algorithm.
- ➔ Clear display of the results in a formatted manner.

Output

KSA 01 Report and Code X

CSE 323 212 D4 KSA-01 T X

CSE323_Project_Report_T X

RSA Encryption

Desktop/New Text Document.html

RSA Algorithm Implementation

Enter First Prime Number:

11

Enter Second Prime Number:

13

Enter Plain Text:

hello

Apply RSA

Results

Public Key (n, e): (143, 7)

Private Key (n, d): (143, 103)

Encrypted Message: 91 62 4 4 45 98

Decrypted Message: hello

KSA 01 Report and Code X CSE 323 212 D4 KSA-01 T X CSE323_Project_Report_T X RSA Encryption

ktop/New Text Document.html

RSA Algorithm Implementation

Enter First Prime Number: 17

Enter Second Prime Number: 19

Enter Plain Text: green university

Apply RSA

Results

Public Key (n, e): (323, 5)

Private Key (n, d): (323, 173)

Encrypted Message: 69 190 271 271 230 223 53 230 22 169 271 190 115 22 165 49

Decrypted Message: green university

Conclusion

This project provides a functional and user-friendly implementation of the RSA algorithm in a web-based application. Users can experiment with different prime numbers and plaintext messages to observe the encryption and decryption process. The application serves as an educational tool to understand the fundamentals of asymmetric cryptography and the practical application of the RSA algorithm in securing data transmission. Further enhancements could include improved error handling, additional features such as key generation, and optimization for performance. Overall, this project demonstrates the versatility and power of the RSA algorithm in ensuring secure communication over the internet.